

Data Flow Diagram & User Stories

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Project Name: Import Data Using Transform Maps in Sheets

Maximum Marks: 4 Marks

Data Flow Diagrams

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It shows where the data comes from, how the data is processed, where it is stored, and how the processed information is provided to the user.

In the **Import Data Using Transform Maps in Sheets** project, the DFD explains the complete flow of importing data from Google Sheets or CSV files into the target system.

Main Data Flow

Google Sheets / CSV File → Upload Data → Transform Map → Field Mapping → Data Validation → Data Transformation → Target System / Database

1. Data Source

The user first prepares the required data in a **Google Sheet or CSV file**. The sheet may contain details such as names, IDs, email addresses, department information, marks, or other required fields.

2. Upload Data

The prepared file is uploaded into the system. The system reads the columns and records from the uploaded file.

3. Transform Map

The Transform Map controls how the source data should be transferred to the target table. It helps connect the fields in the source file with the corresponding fields in the target system.

4. Field Mapping

The administrator maps each source field to the correct target field. For example, a source field called **Student Name** can be mapped to the target field **Name**.

5. Data Transformation

If the source data format is different from the target format, transformation rules are applied. This helps convert the data into the required format before storing it.

6. Data Validation

The imported data is checked for missing values, incorrect formats, invalid records, and other errors. This helps maintain data quality.

7. Error Handling

If any record fails during the import process, the system identifies the error. The administrator can correct the data and process the record again.

8. Duplicate Handling

The system can check for duplicate records before inserting data. This helps prevent unnecessary duplicate entries in the target system.

9. Target System / Database

After successful validation and transformation, the records are inserted into the required target table or database.

10. Import Status

The user or administrator can check whether the import was successful, failed, or partially completed.

DFD Components

Component	Description
User	Prepares the data and uploads the file.
Google Sheets / CSV	Acts as the source of data.
Transform Map	Controls the mapping and transformation of source data.
Field Mapping	Connects source fields with target fields.
Validation	Checks data for errors and missing values.
Error Handling	Identifies and manages failed records.
Target Database	Stores the successfully imported records.
Administrator	Configures mappings and monitors the import process.

DFD Description

The user starts by preparing the required data in Google Sheets or a CSV file. The file is then uploaded into the system. The Transform Map reads the source data and maps each source field to the appropriate target field.

After mapping, transformation rules are applied wherever necessary. The system validates the records to identify missing, invalid, or incorrectly formatted data. Records that pass validation are inserted into the target database.

If any record contains an error, the system displays the error information. The administrator can correct the data and reprocess the failed records. The administrator can also monitor the import status and verify that the records have been stored correctly.

This complete flow reduces manual data entry, saves time, improves accuracy, and makes the data import process easier to manage.

User Stories

User stories describe the requirements of the system from the user's point of view. They explain **who needs the feature, what they want to do, and what result they expect**.

User Story Table

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Data User	Data Preparation	USN-1	As a user, I can prepare the required data in Google Sheets or CSV format before importing it.	The file contains the required columns and valid records.	High	Sprint-1
Data User	Data Upload	USN-2	As a user, I can upload a Google Sheets/CSV file into the system.	The file is uploaded successfully and the system can read the data.	High	Sprint-1
Data User	Data Import	USN-3	As a user, I can import data from the uploaded file without entering every record manually.	Records are imported successfully into the target system.	High	Sprint-1

Data Administrator	Field Mapping	USN-4	As an administrator, I can map source fields to the correct target fields using Transform Maps.	All required source fields are mapped to the correct target fields.	High	Sprint-1
Data Administrator	Field Mapping	USN-5	As an administrator, I can review the field mappings before starting the import.	The administrator can verify and modify mappings before importing.	High	Sprint-1
Data Administrator	Data Transformation	USN-6	As an administrator, I can apply transformation rules when source and target formats are different.	Source values are converted into the required target format.	Medium	Sprint-2
Data User	Data Validation	USN-7	As a user, I can validate the uploaded data before it is inserted into the target system.	Missing and invalid records are identified before insertion.	High	Sprint-2
Data Administrator	Error Handling	USN-8	As an administrator, I can view errors that occur during the import process.	Error messages clearly identify the failed records and reasons.	High	Sprint-2

Data Administrator	Error Handling	USN-9	As an administrator, I can correct failed records and reprocess them.	Corrected records can be successfully imported.	Medium	Sprint-2
Data Administrator	Duplicate Handling	USN-10	As an administrator, I can identify duplicate records during the import process.	Duplicate records are identified and unnecessary duplicates are prevented.	Medium	Sprint-3
Data User	Import Status	USN-11	As a user, I can view the status of the data import process.	The system displays successful, failed, and pending records.	Medium	Sprint-3
Data Administrator	Import Monitoring	USN-12	As an administrator, I can monitor the overall import activity.	Import activities and results can be reviewed.	High	Sprint-3
Data Administrator	Data Verification	USN-13	As an administrator, I can verify that imported data is stored in the correct target fields.	Imported records match the expected target field values.	High	Sprint-3
Data User	Data Accuracy	USN-14	As a user, I can import data accurately	Imported data is available correctly in	High	Sprint-3

			without manually entering every record.	the target system.		
Administrator	Import Summary	USN-15	As an administrator, I can view a summary of successful and failed records after import.	The system displays a clear import summary.	Medium	Sprint-4
Administrator	Data Management	USN-16	As an administrator, I can manage and update Transform Map configurations when requirements change.	Mapping configurations can be updated and saved successfully.	Medium	Sprint-4
Administrator	Security	USN-17	As an administrator, I can control who is allowed to configure Transform Maps and import data.	Only authorized users can perform administrative activities.	High	Sprint-4
Data User	User Experience	USN-18	As a user, I can understand the import process easily without complex manual steps.	The import process is simple and understandable to the user.	Medium	Sprint-4

Detailed User Stories

USN-1 – Data Preparation

User Story:

As a user, I can prepare the required data in Google Sheets or CSV format before importing it.

Acceptance Criteria:

- Required columns are available.
- Data is entered in the correct format.
- Empty or unnecessary columns can be identified before upload.

USN-2 – Data Upload

User Story:

As a user, I can upload a Google Sheets or CSV file into the system.

Acceptance Criteria:

- The file can be uploaded successfully.
- The system recognizes the uploaded file.
- Source columns and records are displayed correctly.

USN-3 – Data Import

User Story:

As a user, I can import data from the uploaded file without entering every record manually.

Acceptance Criteria:

- Data is transferred automatically.
- Manual data entry is reduced.
- Imported records are available in the target system.

USN-4 – Field Mapping

User Story:

As an administrator, I can map source fields to the correct target fields using Transform Maps.

Acceptance Criteria:

- Source fields are displayed.
- Target fields are available.
- Each required field can be mapped correctly.
- Incorrect mappings can be changed.

USN-5 – Review Field Mapping

User Story:

As an administrator, I can review the field mappings before starting the import.

Acceptance Criteria:

- Mapping configuration can be viewed.
- Incorrect mappings can be corrected.
- Administrator can confirm the mapping before import.

USN-6 – Data Transformation

User Story:

As an administrator, I can apply transformation rules when source and target formats are different.

Acceptance Criteria:

- Transformation rules are applied correctly.
- Data follows the target format.
- Transformed values are stored correctly.

USN-7 – Data Validation

User Story:

As a user, I can validate the uploaded data before it is inserted into the target system.

Acceptance Criteria:

- Missing values are identified.
- Invalid values are identified.
- Incorrect formats are reported.
- Valid records can continue to the import process.

USN-8 – Error Handling

User Story:

As an administrator, I can view errors that occur during the import process.

Acceptance Criteria:

- Failed records are identified.
- Error reasons are displayed.
- Administrator can understand what needs to be corrected.

USN-9 – Reprocess Failed Records

User Story:

As an administrator, I can correct failed records and reprocess them.

Acceptance Criteria:

- Failed records can be corrected.
- Corrected records can be submitted again.
- Successfully corrected records are imported.

USN-10 – Duplicate Handling

User Story:

As an administrator, I can identify duplicate records during the import process.

Acceptance Criteria:

- Duplicate records can be identified.
- Duplicate entries are prevented where applicable.
- Existing records are not unnecessarily duplicated.

Benefits of the Proposed System

1. Reduces Manual Work

The system reduces the need to manually enter every record from a spreadsheet into the target system.

2. Saves Time

Large amounts of data can be imported more quickly compared with manual data entry.

3. Improves Data Accuracy

Field mapping and validation help reduce incorrect data entry and mapping mistakes.

4. Easy Data Transformation

Transform Maps allow data to be changed into the required format before it is stored.

5. Better Error Identification

The system can identify records that fail during the import process and provide error information.

6. Reduces Duplicate Records

Duplicate checking helps maintain cleaner and more organized data.

7. Easy Monitoring

Administrators can monitor the import status and check successful and failed records.

8. Supports Large Data

The approach is useful when the spreadsheet contains a large number of records.

9. Improves Data Management

The administrator can control field mappings, transformation rules, validation, and import activities.

10. Simple Import Process

The overall process makes spreadsheet-to-system data transfer easier and more organized.

Overall Project Flow

Prepare Data in Google Sheets/CSV

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Upload the File

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Read Source Data

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Configure Transform Map

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Map Source Fields to Target Fields

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Apply Transformation Rules

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Validate Data

↓

Check Errors and Duplicates

↓

Import Valid Records

↓

Store Data in Target Database

↓

View Import Status and Summary